

Data Beyond Google Trends

Wikipedia pageviews and wastewater, scraped with an agent

Shihao Yang • Candice Djorno

July 21, 2026

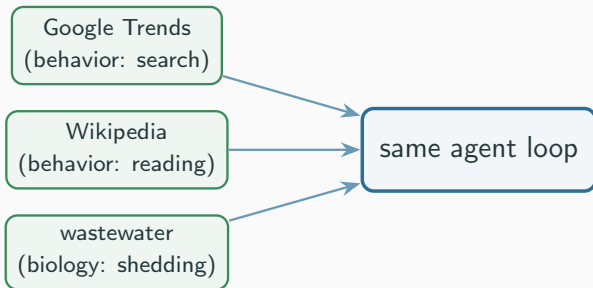
SISMID 2026 • Day 2, 9:00

Why more than search

Yesterday you got fluent with **one** stream. Google Trends is powerful but has real blind spots: attention bias, English skew, sampling instability.

Today

Add two streams that fail *differently*, so their errors do not line up: **Wikipedia pageviews** and **wastewater**. Same agent workflow, two new sources.



Google Trends revisited: the API

Wikipedia pageviews

Wastewater surveillance (CDC)

Continuing the Day 1 workflow

Google Trends revisited: the API

First, let us fix yesterday

Before adding new streams, we repair the one you already know.

- Yesterday you scraped Google Trends with `pytrends` and met its two problems: **HTTP 429** rate limits, and values that **change between pulls**, because the public endpoint is *sampled*.
- There is an official **Google Trends API**. It returns the **same normalized 0–100 index**, so nothing about your analysis changes.

	pytrends (scraping)	Trends API
Access	unofficial endpoint	documented API, needs a key
Failures	HTTP 429 roulette	none in normal use
Values	resampled each pull	identical every pull
Signal	normalized 0–100	the same 0–100

Unlock the class key

In the terminal, before you start Jupyter

```
source scripts/unlock-gt-api-key.sh
```

Exports GT_API. If Jupyter is open, **restart the kernel**.

```
GET https://www.googleapis.com/trends/v1beta/graph
?key=$GT_API
&terms=dengue&terms=/m/0cycc # terms OR topic ids
&restrictions.geo=MX # US, MX, US-GA ...
&restrictions.startDate=2021-07 # YYYY-MM
&restrictions.endDate=2026-07
```

- **Quota: 1,000 requests/day for the whole room**, health topics only.
- Window length picks weekly vs monthly, exactly as on the website.

Term vs topic: why today needs topics

A **term** is the literal string. A **topic** is Google's Knowledge Graph entity (`/m/0cycc` = Influenza), folding together every spelling, synonym and **language**.

Country	term "flu"		topic /m/0cycc	
	max	mean	max	mean
United States	88	22.1	100	24.9
France	3	0.9	100	16.3
Italy	4	1.6	100	17.7
Mexico	4	1.7	100	37.1

An English term is nearly silent abroad. One topic id works in every language, which is what makes going multi-country today possible.

► **Open** `notebooks/00_google_trends_api.ipynb` and pull your own topic.

What the API does not fix

The API makes the *delivery* reliable. It does not change what search *is*.

- Google Trends still measures **attention**, not infection: news coverage moves it too.
- Low-volume places still return a lot of **zeros**.
- The relationship to disease still **drifts** between seasons.

So we still need other streams

A reliable pipe does not make one signal sufficient. Next: streams that fail *differently*, so their errors do not line up.

Wikipedia pageviews

What Wikipedia pageviews are

A free, hourly log of how many times each article was viewed, exposed through one official **Wikimedia REST API** (no key).

- Views of “Dengue fever” or “Influenza” track public attention, much like search, but from a **different behavior**: reading, not querying.
- **Stable and reproducible**: pull the same series twice and the numbers match. A welcome contrast to Google Trends.
- Deep history (2015 onward), consistent methodology, well documented.

A dependable, well-behaved stream. Good to learn the multi-source workflow on.

Language is a feature

Wikipedia is per-language, so one concept gives you **several** signals at once.

- The **Spanish** article speaks for much of Latin America, **Portuguese** for Brazil, **English** for global attention.
- This is where Wikipedia beats Google Trends: strong **country-level** and **non-English** signal, exactly where search coverage is thin.
- A natural extra time series for the Dengue exercise: overlay the Spanish and Portuguese dengue articles on the case curve.

Pick the language that matches the population you are trying to see.

Scrape it with the agent

Same five-step loop as yesterday. Only the fetch changes.

Prompt

```
Using the Wikimedia REST pageviews API (no key, set a
descriptive User-Agent), pull daily views for the
"Dengue fever" article in English, Spanish, and
Portuguese since 2016. Use all-access and all-agents,
resolve redirects to the canonical title, align to
weekly, merge on date, plot the three, and report the
date range and any gaps.
```

Then the habit: one plot, one sanity question, before it enters a model.

- **Article titles differ by language** and can be renamed; a wrong title silently returns nothing. Resolve to the canonical page and check the count is non-zero.
- **Access and agent filters matter:** all-access vs mobile/desktop, and all-agents vs user (exclude bots) change the series.
- **Attention spikes:** a news event or a linked front-page item can spike views with no change in disease, just like search.
- **Disambiguation:** “Cold” the illness vs the temperature. Prefer specific articles or topics.

Wastewater surveillance (CDC)

A fundamentally different kind of signal

Wikipedia and Google Trends measure **behavior**. Wastewater measures **biology**.

- Treatment plants are sampled for pathogen genetic material (viral RNA). It counts **shedding** by infected people, whether or not they search, test, or seek care.
- Because people shed early, it can **lead** clinical reporting, and it captures mild and asymptomatic cases that never reach a clinic.
- Publicly available from the CDC **National Wastewater Surveillance System (NWSS)**, by site and by jurisdiction.

The one stream here that does not depend on human attention at all.

Scrape it with the agent

Prompt

```
From CDC's National Wastewater Surveillance System (NWSS)
open data on data.cdc.gov, pull the weekly wastewater
viral activity level for Georgia (state aggregate, and
list the reporting sites). Tidy to date + value, plot it,
and overlay a reference such as ILI or ED visits so I can
see whether it leads. Report site coverage and any gaps.
```

The agent finds the dataset, handles the query, and returns a tidy series; you judge whether it actually leads.

Wastewater gotchas

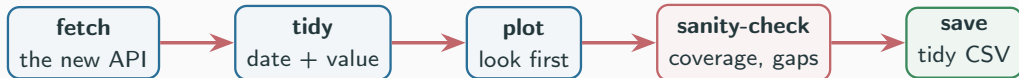
- **Coverage varies:** which plants report, and what fraction of the population they cover, changes over time and by state. Gaps are common.
- **Normalization:** raw concentration is adjusted for flow and for a fecal marker (PMMoV). Compare like with like; do not mix raw and normalized.
- **Single site is noisy:** a regional or state aggregate is far more stable than one plant.
- **Reporting lag and cadence:** it is fast, but not instant, and not every site reports every week.

Assess it honestly

The claim is early warning. Check the **lead time against ILI** yourself, as in Santillana's wastewater preprint, rather than assuming it.

Continuing the Day 1 workflow

Same loop, same habits, now three streams



- Nothing new to learn in the *workflow*: describe the outcome, let the agent fetch, then verify. Only the source changed.
- Wikipedia and CDC are well-behaved (no datacenter block), so retries and the proxy rarely come up here. Google Trends was the hard one.

You now hold a small multi-stream bundle

For your disease and place, you can now assemble, side by side:

- Google Trends (search behavior), preprocessed,
- Wikipedia pageviews (reading behavior, multi-language),
- wastewater (biological shedding),
- all against a reference such as ILI or ED visits.

This bundle is the raw material for the capstone: independent signals whose errors do not line up.

Looking ahead

- **Next (11:00):** more novel streams, human mobility (ground and air) and news alerts for emerging outbreaks.
- **This afternoon:** fold these signals into **ARGO**, alongside the disease's own history.
- **Day 3:** bring your own problem, and combine whichever streams fit it.

Two more streams down, the same loop each time. The library of signals is growing.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`